

2019 Ph H2 Q5

Section: Our Dynamic Universe

Topic: Relativity

A spacecraft moves away from Earth at a speed of $0.80c$. A clock on the spacecraft measures the time interval between two events as 12 s .

(a)
Calculate the time interval between the two events as measured by an observer on Earth

Use the **time dilation** formula:

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where:

- $t_0 = 12\text{ s}$
- $v = 0.80c$

$$t = \frac{12}{\sqrt{1 - (0.80)^2}} = \frac{12}{\sqrt{1 - 0.64}} = \frac{12}{\sqrt{0.36}} = \frac{12}{0.6} = 20\text{ s}$$

(b)
Explain why the observer on Earth measures a longer time interval

✔ According to **special relativity**, time appears to run **more slowly** on a moving spacecraft when observed from Earth. This effect is called **time dilation**.
The moving clock records less time between the two events, so the stationary observer (on Earth) measures a **longer** time interval.

🧠 Revision Tips

- Use $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ for time dilation:
 - t : time measured by the **stationary** observer
 - t_0 : proper time (measured in the moving frame)
- If you're on the spacecraft: you experience **proper time**
- If you're observing the spacecraft from Earth: you see **dilated time**